Economics 5040: Dynamic Stochastic Modeling
Midterm Study Guide

Fall 2026

The midterm will be **closed book and closed notes**. The emphasis will be on understanding and explaining the central economic ideas of the course. You should be able to explain each of the following topics clearly and concisely in your own words.

1. Inter-temporal Prices

- Understand how the real interest rate determines the relative price of consumption today and consumption in the future.
- Be able to explain the substitution incentive created by a change in the real interest rate.

2. Choice Under Uncertainty

- Understand expected utility, risk aversion, and how the curvature of the utility function affects the value of a risky prospect.
- Be able to compute and interpret the certainty equivalent of a simple gamble.

3. Markov Processes

- Understand the meaning of the Markov property: conditional on the current state, the past provides no additional information about the distribution of the next state.
- Know how to interpret a transition matrix P, including the economic meaning of an element P_{ij} and of an entire row.

4. Introduction to Dynamic Programming

- Understand the Bellman equation as a decomposition of a dynamic problem into a current reward and a discounted expected continuation value.
- Be able to identify the state variable, choice variable, current payoff, and next-period state in a dynamic economic problem.

5. Value Functions and Policy Functions

- Understand the distinction between a value function and a policy function.
- Know what economic information each function provides and how the optimal policy is related to the Bellman equation.

6. Howard's Method versus Value Function Iteration

- Understand the basic logic of value function iteration: repeatedly applying the Bellman operator until the value function converges.
- Understand how Howard's method separates policy evaluation from policy improvement, and why this can make computation much faster.

7. The Solow Model

- Understand how saving and capital accumulation determine the steady-state levels of capital and output.
- Be able to distinguish a change in the *level* of steady-state output from a permanent change in the long-run growth rate.

8. Optimal Saving

- Understand the Euler equation

  $$u'(c_t) = \beta R E_t[u'(c_{t+1})]$$

  as a comparison between the marginal value of consuming today and the expected discounted marginal value of saving for tomorrow.
- Be able to explain how a household changes consumption and saving when the two sides of the Euler equation are not equal.

9. Income Fluctuations

- Understand precautionary saving: uncertainty about future income can increase desired saving even when expected future income is unchanged.
- Understand why assets provide self-insurance by transferring resources toward future states in which consumption and wealth are low.

10. Putting the Ideas Together: Job Search

- Understand why job search is a dynamic decision problem: today's decision affects both today's payoff and the opportunities available tomorrow.
- Be able to formulate a job-search problem in terms of state variables, a choice variable, the transition to tomorrow's state, and a continuation value.

How to prepare. Do not concentrate on memorizing long derivations or Python code. For each topic above, make sure that you can explain the central idea in a short paragraph, interpret the important equations economically, and identify how the pieces of the model fit together.